

soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Data Managers

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DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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1. Overview

This manual has been developed as a reference guide to assist Med-SIS STAFF and Data Managers in navigating and using the Med Soil Information System (Med SIS).

The SIS is essentially a **Web GIS Catalogue** of resources related to soil data. It is based on Geonode (<https://geonode.org/>), whose resources include datasets, maps, documents, geostories and dashboards.

In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool, known as the **Back Office**

MED-SIS staff and data managers use the Back Office to create and configure catalogue resources and to manage them through the Web GIS catalogue.

This document is divided in three main sections: “Web GIS Catalogue”, “Back Office”, “Admin panel”

Further details on Web GIS Catalogue functionalities can be found on the official GeoNode documentation page at <https://docs.geonode.org/en/latest> by selecting the “User Guide” menu

2. Web GIS Catalogue

Within the SIS, only a limited subset of content is publicly accessible. Registered generic users therefore have read-only access and can interact with resources only in accordance with the permissions explicitly defined by the data manager responsible for each dataset.

The data manager is responsible for data ingestion into the Med-SIS and **defines the access control policies** applied to each dataset, **based on the intended users and their roles**. The data manager can add datasets and documents, create maps, dashboards and geostories, and edit and share resources with other users.

Med-SIS homepage

The Med-SIS system has a moderate user registration by default; this means that the users must be validated by the staff after the registration (Figure 6)

Figure. 2 Sign in page



Figure. 3 Med-SIS homepage after the login

As shown in Figure 3 the system's main page displays the project logo in the upper-left corner. Clicking the logo redirects the user to the homepage .

In the upper-right corner, there is the language selection menu and the user menu which provides access to user-related actions, as illustrated in Figure 3.

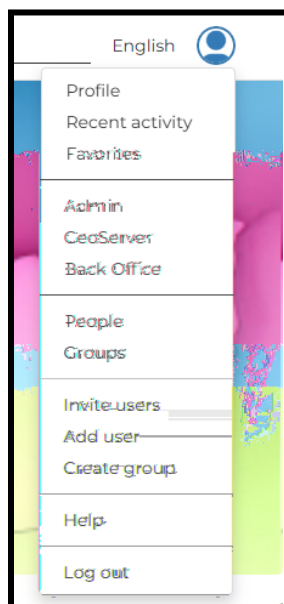


Figure 4

- **Profile:** displays the user's profile
- **Recent activit :** displays the user's recent activities
- **Favorites:** displays the list of the favorites datasets
- **Admin:** opens the admin panel **if the user is a member of the staff**
- **Geoserver:** opens the Geoserver application
- **Back Office:** open the back office **if the user is a member of the staff**
- **People, Groups, Invite user, Add User, Create Group:** Users and groups management - **only if the user is a member of the staff**
- **Help:** provides quick help on performing actions in the catalogue
- **Log out:** logs the user out of the system.

In the top area of the MED-SIS homepage, there are menus providing direct access to the available resources, organized by category (Datasets, Maps, Documents, Geostories, and Dashboards) or collectively under All Resources.

The same resources can also be accessed by scrolling down the page.

In the center of the page, there are quick links providing direct access to the following resources:

- **Soil Maps** : Vector / Raster datasets containing Pedological Classification
- **Soil Point Data** : Point vector datasets containing soil profile and samples data
- **Chemical Health** : Vector / Raster datasets containing soil indicators describing chemical health
- **Physical Health**: Vector / Raster datasets containing soil indicators describing physical health
- **Biological Health**: Vector / Raster datasets containing soil indicators describing biological health
- **Other Maps**: Vector / Raster datasets containing generic gis data
- **Help Manuals**: MED-SIS manuals in pdf format like this data

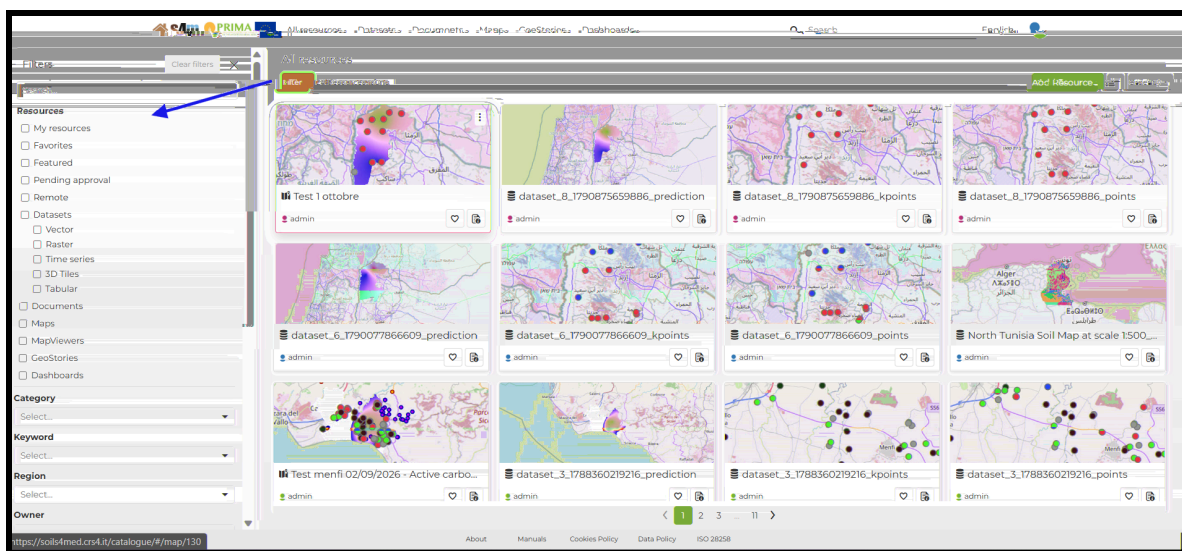


Figure. 5 Resources' List and Filters

Once the list of available resources is displayed (Figure. 5);, resources can be filtered using one or more of the following filters:

- **Featured**: resources explicitly marked by data managers or administrators as highlighted or recommended
- **Remote**: resources served from an external service, such as a Web Map Service (WMS), an ArcGIS REST service, or another GeoNode instance
- **Dataset**: filters datasets by type including Vector, Raster, Time series, and 3D Tiles
- **Documents**: filters the list to show available documents
- **Maps**: filters the list to show available maps including maps created using the Geonode Map Viewer, maps created through MapStore-based viewer configuration files, maps created via API, or scripts, and maps imported from other Geonode instances
- **MapViewers**: filters the list to show only maps created using the Geonode Map Viewer

- **GeoStories:** filters the list to show available Geostories
- **Dashboards:** filters the list to show available Dashboards
- **Category:** filters resources based on the value assigned to this metadata field. Typical categories, often based on INSPIRE / ISO topic categories used in geospatial metadata standards, include Boundaries, Environment, Transportation, Economy, Health, Society, and Utilities / Communication. MED-SIS also includes **Point Soil Data Chemical Health Physical Health** and **Biological Health**.
- **Keywords:** filters resources based on the keywords assigned in the metadata, typically free text tags. Several Med-SIS keywords have been added such as “Pedological map”
- **Regions:** filters resources based on the geographical region specified in the metadata. This is different from the bounding box, which is automatically derived from the geographical extent of the data.
- **Owner:** filters resources based on the user who created or owns them
- **Group:** filters resources associated with a specific GeoNode group
- **Date from:** filters resources based on their creation or upload date, showing only resources added on or after the specified date
- **Date to:** filters resources based on their creation or upload date, showing only resources added on or before the specified date
- **Extent:** filters resources based on their geographical extent, including resources whose geographical area intersects or is contained within the selected extent

User Profile

Once registered, the SIS administrator assigns new members the Data Manager role in accordance with the system’s role-based access control policies.

Communication among the different actors within the SIS can take place through the platform’s internal messaging system or via private email.

By selecting the “Profile” option from the user menu, the data manager is directed to the profile management interface, where they can:

- configure personal information by clicking the *Edit Profile* button. Users can update their profile by adding a profile photo, specifying their organization, writing a description, indicating their position, and setting their location
- manage email settings by clicking the *Associates emails* button
- manage credentials such as the password, by clicking the *Set/Change password* button
- access a summary of their activities in the SIS by clicking the *My Activities* button, which displays the user’s latest actions within the system
- configure notification settings, including how and when to receive notifications and for which events, concerning activities on the platform by other users or administrators, by clicking the *Notification* button
- creating announcements

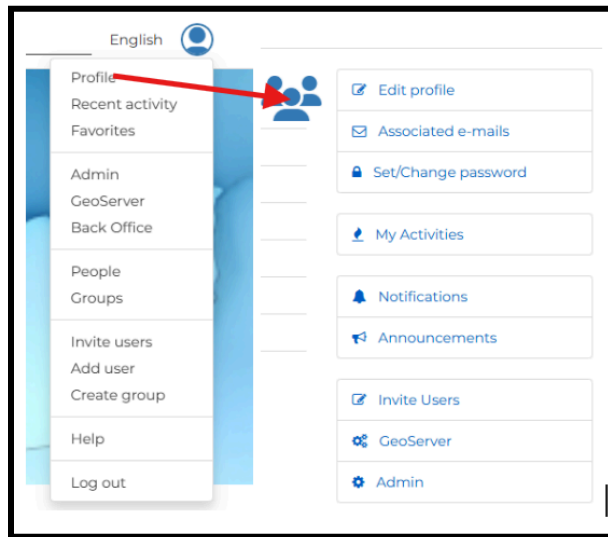


Figure 5 User profile menus

The **Recent Activity** option provides a detailed overview of the actions performed within the SIS by active users. It allows users to view the type of activities performed and identify the users who performed them.

The **Favorites** option provides access to the list of resources marked as *favorites*, allowing users to easily retrieve them without having to search for them in the catalogue.

The **Inbox** option is part of the Geonode internal messaging system and is designed to facilitate communication between registered users and group members. It acts as a hub for receiving, reading, and replying to messages within the SIS platform.

The other options in the menu provide access to administrative tools such as **Admin** and **Geoserver**, the **Back Office** for uploading and managing soil data, the **Help** section, which provides guidance on using GeoNode functionalities and the **Log out** button.

The *Admin Panel* and *Back Office* are described in the respective sections of this manual. Geoserver, which is responsible for publishing and serving the spatial data of the SIS platform through standard OGC web services such as WMS, WFS, and WCS, should not be used by data managers and is intended for use by the system administrators only.

Resources

The data manager can manage all existing catalogue resources, which are organized into the following categories:

Dataset

Datasets are geospatial resources that can be stored, managed, visualized, and shared through the platform. A dataset typically represents a spatial layer.

The Dataset view has several menus and options for managing and accessing the resource.



Figure 6 Resource menu

By clicking on the **Open Property button**, the user can view the resource's general information, geographic location, alphanumeric attributes, and linked resources that use or are used by the dataset.

By clicking the **Resource menu**, users with staff permissions can save or delete the resource.

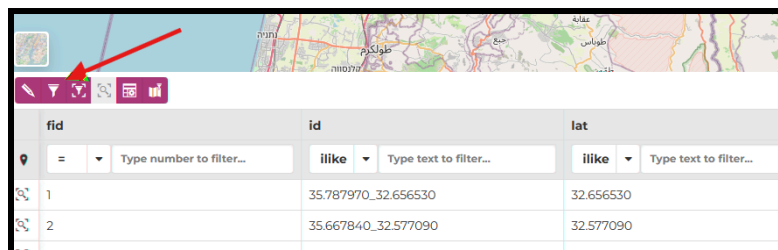


Figure 7 View Data menu

By clicking the **View menu**, the user can access the resource's metadata (raster and vector) or the alphanumeric data associated with the dataset (only vector). When "view data" is selected, the advanced search icons in the upper left corner of the table (Figure 7) provide the following options:

- perform an advanced search including attribute filtering, area of interest selection and layer filtering;
- filter the data displayed in the table according to the current view, after zooming to the desired area;
- zoom to page extent;
- hide or show columns in the table;
- synchronize the map with the filters applied to the data table.

Examples of queries using the "advanced search" button are shown below.

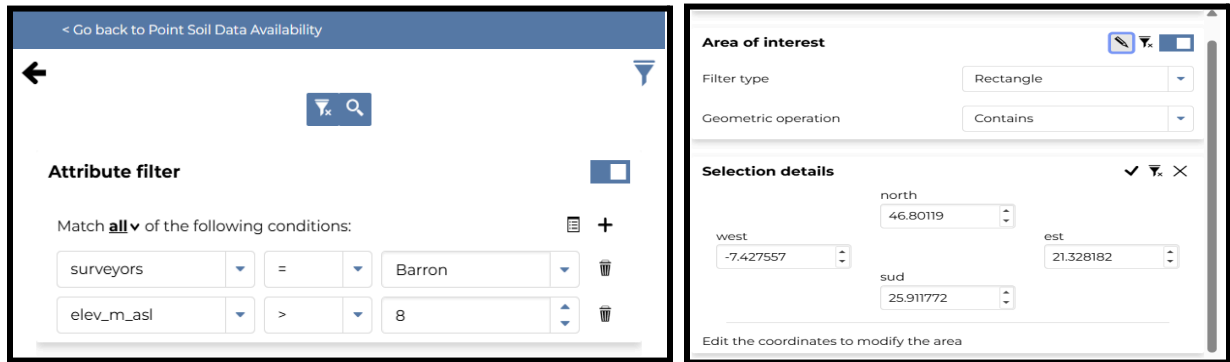
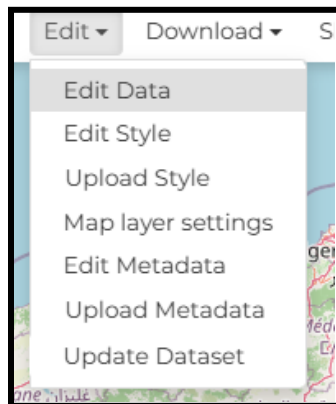


Figure 8 filters in the view data tool

By clicking the **Edit menu** (Figure 8), the user can edit dataset properties and perform the following operations:

- **Edit data:** opens the tool for editing the data table, if available and if the user has



The “viewport” filter limit the search results d to resources that fall within the current map extent i.e. the visible area displayed on the screen.

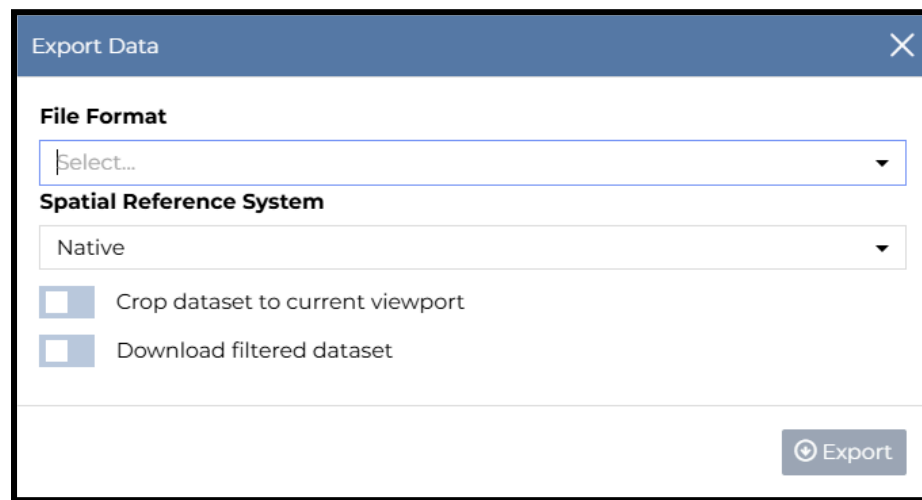
A dialog box titled "Export Data" with a close button (X) in the top right corner. It contains two sections: "File Format" with a dropdown menu showing "Select..." and "Spatial Reference System" with a dropdown menu showing "Native". Below these are two checkboxes: "Crop dataset to current viewport" and "Download filtered dataset", both of which are currently unchecked. At the bottom right is a button labeled "Export" with a download icon.

Figure 10: Download form

By clicking the **Share menu**, users with “Manage” s permissions can configure access permissions for the dataset. Using the permissions settings widget, they can define the following permissions for other users: "View", "Download", "Edit", or "Manage".

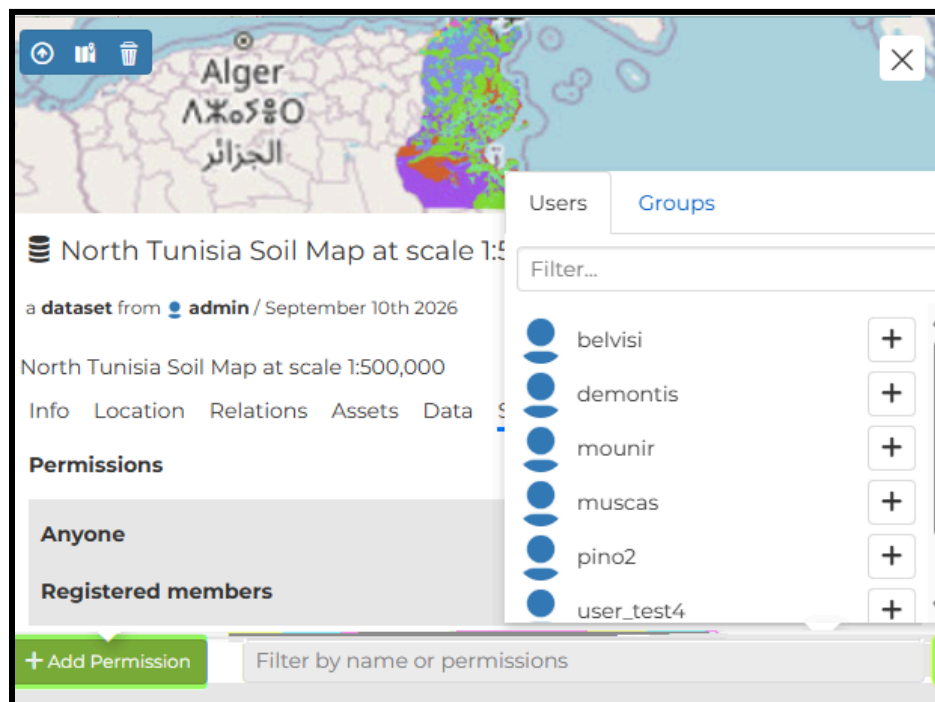
A screenshot of a web application showing a map of North Tunisia with a soil map overlay. A modal window titled "Permissions" is open, displaying a list of users and groups. The "Users" tab is selected, showing a list of users: belvisi, demontis, mounir, muscas, pino2, and user_test4. Each user has a blue circular icon and a "+" button next to it. The "Groups" tab is also visible. Below the list is a search bar labeled "Filter by name or permissions". At the bottom left is a green button labeled "+ Add Permission".

Figure 11: Permissions settings widget

To protect the dataset, it is recommended to set the permission level to "none" for anonymous and registered users. Where appropriate, it is also more secure to assign permissions to individual users rather than to groups.

By clicking the **Filter menu**, the user can perform the same operations available through the icons in the **View menu**. The user can perform advanced searches including attribute-based filtering, and area of interest selection. For attribute filtering, multiple conditions can be applied by adding new conditions using the "+" button.

By clicking the **Create Map link**, the user can create a new **Map resource** using the selected dataset

By clicking the **print tool** the user can configure and generate a pdf

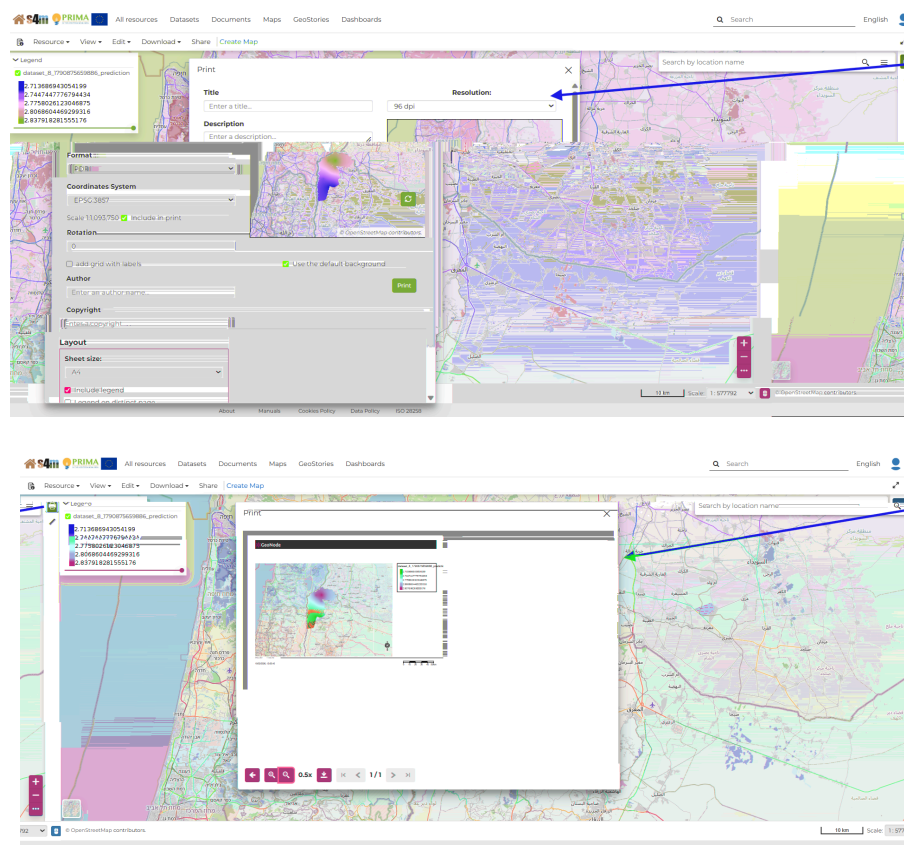


Figure 12: Print widget

Maps

Maps are compositions or visualizations built from *one or more datasets* (layers).

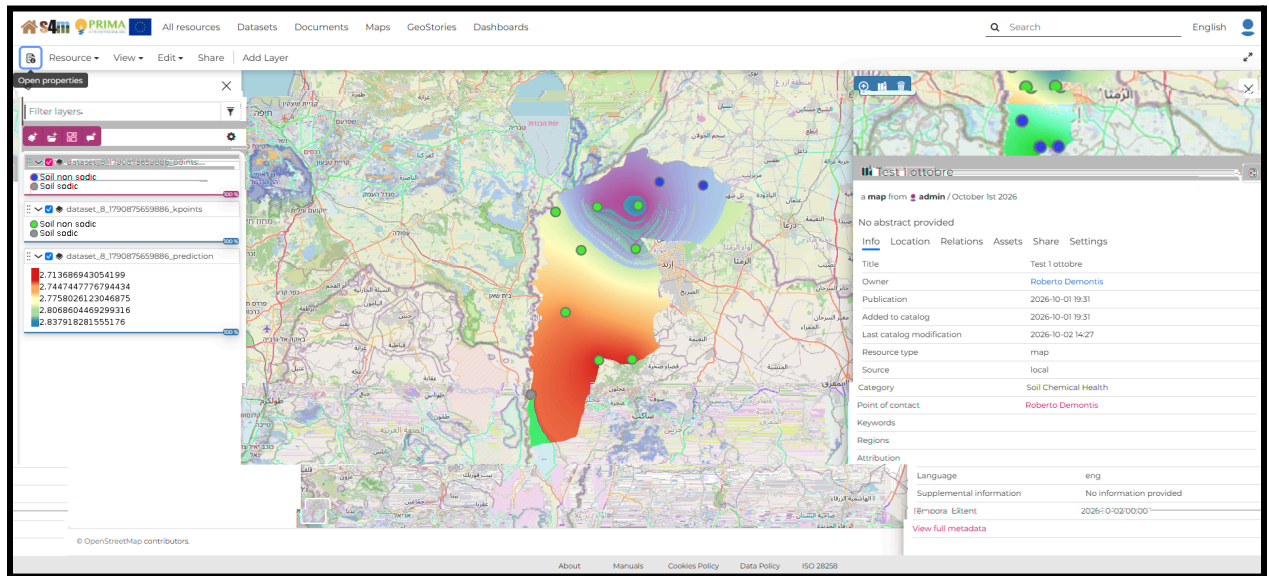


Figure 13, Map resource panel



Figure 14 Map Resource menu

If the user has permissions to view a map and its associated datasets, they can perform the following actions:

by clicking the **Open Property button**, the user can view the resource's general information, geographic location, alphanumeric attributes, and linked resources that use or are used by it.

By clicking the **Resource menu**, users with staff permissions can save or delete the resource.

By clicking the **View menu**, the user can view the resource's metadata.

By clicking the **Edit menu**, the user can edit the map's metadata, which is particularly important for filtering resources or adding viewers.

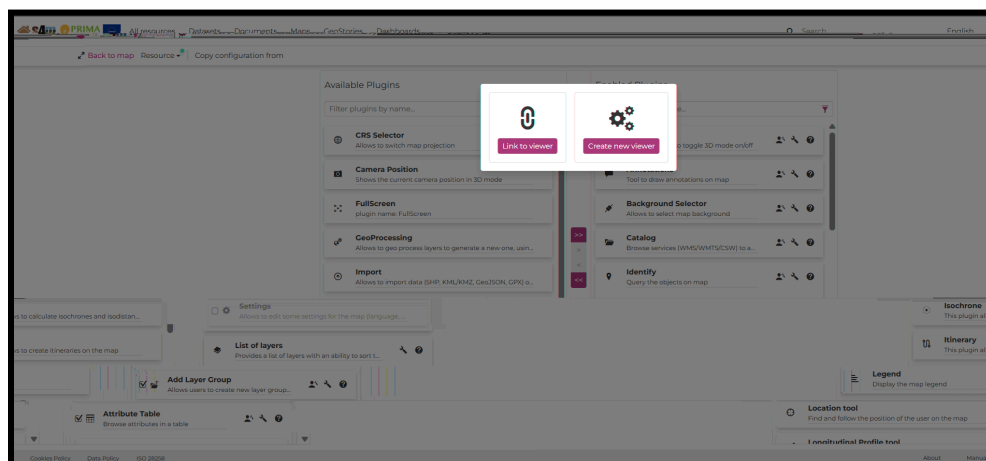


Figure 15, add viewer panel

By clicking the **Share menu**, users with “Manage” permissions can configure access permissions for the dataset. Using the permissions settings widget, they can define the following permissions for other users: "View", "Download", "Edit", or "Manage".

By clicking the **Add Layer link**, the user can add datasets to the map. Datasets can be selected from the catalogue or added from external datasets. External datasets can be configured by clicking the "+" button in the interface.

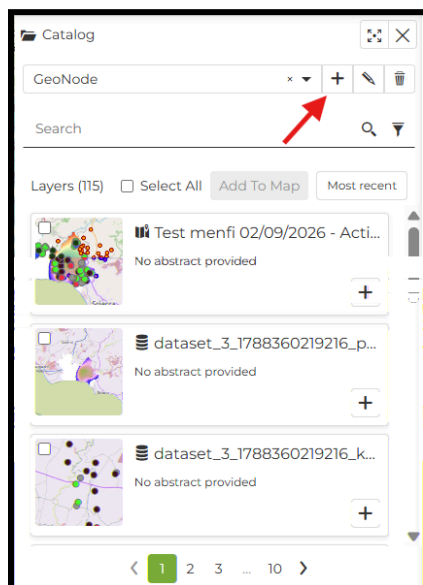


Figure 16, add external layer panel

The panel for exploring and configuring the map is shown in Figure 17

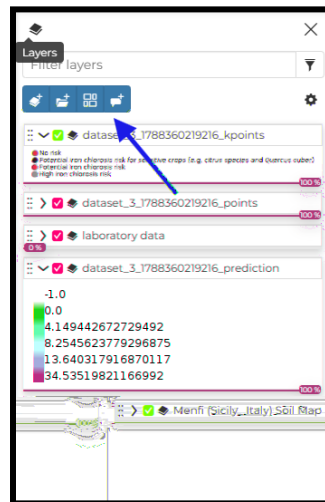


Figure 17, Map properties panel

By clicking the **first** of the four buttons on the left, data managers can add an external layer by selecting one of the following service types: WMS (Web Map Service), WMTS (Web Map Tile Service), CSW (Catalog Service for the Web), TMS (Tile Map Service), XYZ Tiles, WFS (Web Feature Service), and ArcGIS REST

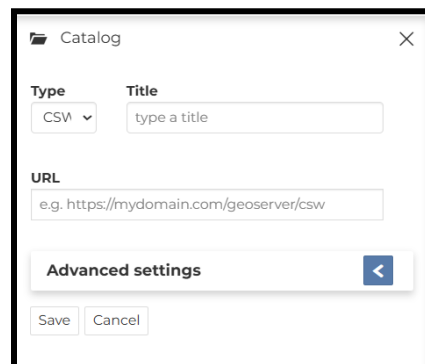


Figure 18, Add layer panel

The Advanced settings control how a remote layer is loaded, displayed, optimized, and interacted with:

- **Loaded:**
 - service search: automatically retrieves the available layers from the selected remote service
 - server type: defines the type of remote server, such as GeoServer or ArcGIS, to ensure that the appropriate requests are sent.
- **Displayed:**
 - preview: (displays a preview of the layer before it is added it
 - format: specifies the image format, such as PNG or JPEG, affecting image quality and

- transparency
- single tile vs tiled: determines whether the map is requested as a single large image rather than as multiple tiles. Single tile mode can provide better for quality but may result in slower performance
- tile size: specifies the tile size, such as 256×256 pixels, affecting performance and the number of requests
- **Optimized:**
 - visibility limits: controls the zoom levels at which the layer is visible
 - tile grids: uses the tiling scheme defined by the remote service to ensure better alignment
 - domain aliases: specifies alternative domains for the same service, which can be useful for load balancing or browser related limitations
- **Interacted with:**
 - interactive legend: allows users to interact with the legend, for example by toggling styles or classes
 - information sheet: defines the format of the feature information displayed when a user clicks on the map, such as HTML, JSON.

By clicking the **second** of the four buttons on the left, users can add a new group of features to the current map.

By clicking the **third** and **fourth** buttons on the left, users can add annotations and widgets to the current map.

Users can typically add the following types of annotations: Points (markers), Lines, Polygons, Text labels / notes.

Users can customize annotation settings, including color, style (such as line thickness, fill, and icon), label and description.

If users have edit permissions, the annotations become part of the map. For generic users, annotations are usually temporary and are lost when the page is refreshed.

Documents

Documents are used to attach and share non-spatial files uploaded to the platform, such as: PDFs (reports, studies), Word documents, Excel spreadsheets, images, and presentations.

Users can open documents, for example by using PDF preview, and download them, depending on their permissions.

As with other SIS resources, documents may be opened, downloaded, edited, and managed according to the user's permissions. However generic SIS users can usually only:

- **View** the document and its metadata
- **Download** the document and the its ISO metadata

Geostories

GeoStories are a special content type used to create interactive narratives that combine maps and multimedia content. They are essentially web-based storytelling pages integrated: maps already available in the SIS, text such as descriptions and explanations), images and other media, and embedded content.

If the user has permissions to view the resources included in a geostory, they can:

- **view** the geostory and its metadata
- **scroll** through its sections/slides
- **read** text and view media content
- **interact** with embedded maps
- **explore** geographic content
- **follow** the narrative flow by navigating between sections, slides, or blocks.

Dashboards

Dashboards are interactive pages used to visualize and analyze geospatial data through charts, widgets, and maps. They are data visualization tools that combine: maps, charts (bar, pie, line, etc.), indicators/counters, filters, and widgets. Their purpose is to summarize and explore data in a dynamic and visual way.

If the user has permissions to view the resources included in a dashboard, they can:

- **View** the dashboard and its metadata, including information about charts, maps, and widgets
- **Interact** with visualizations
- **Apply filters** and view the corresponding dynamic changes
- **Explore** linked maps

If the user has permissions to edit and manage permissions, they can also add and configure widgets

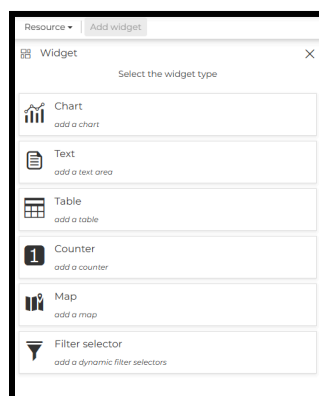


Figure 17, Add widget panel

3. Back Office

In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool known as the "**Back Office**."

Soil data are managed through the MED-SIS Back-office and published in the MED-SIS Catalogue

The Back Office is accessible only to data managers and SIS staff

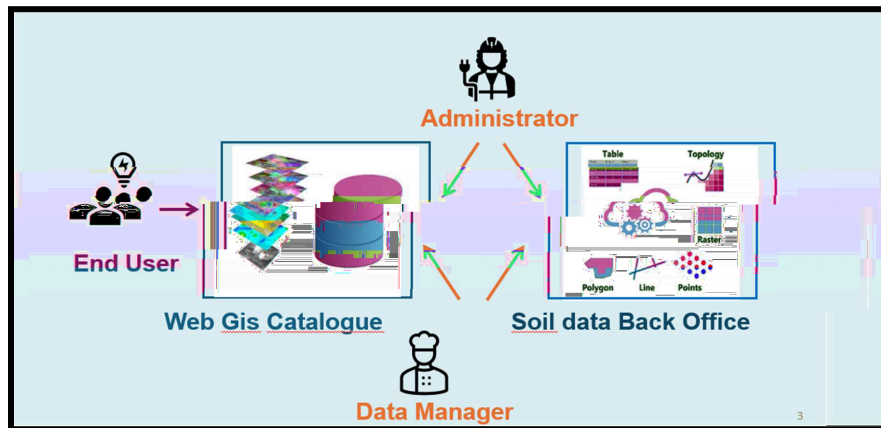


Figure 18, MED-SIS schema

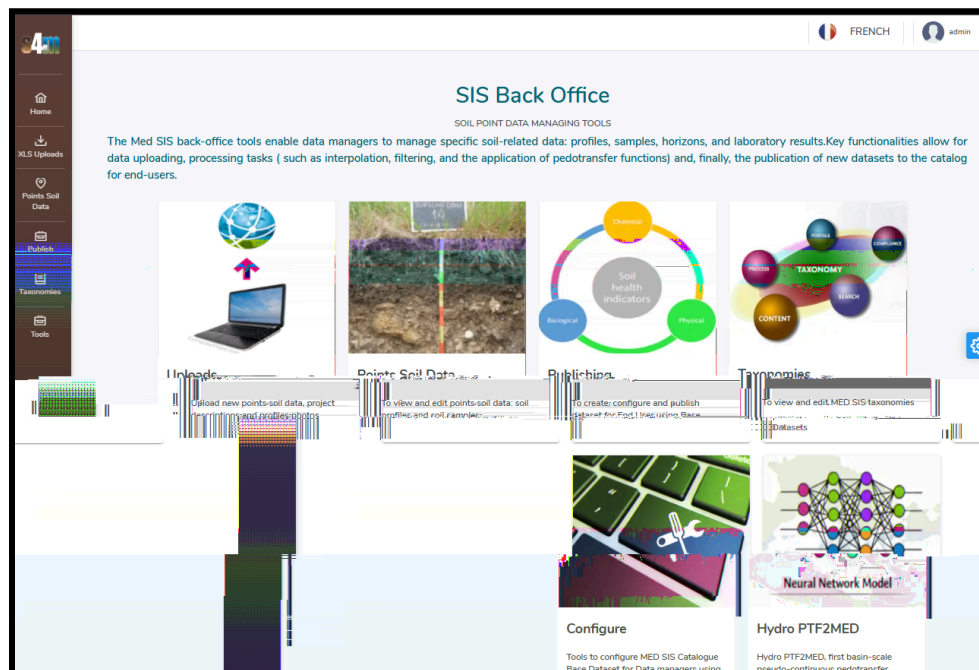


Figure 19, Back Office home page

Data Managers use Back Office tools to manage, filter, and process **base datasets** in order to create and publish **customized datasets for end users**.

The Back Office homepage (Figure 19) displays the available tools. Each tool can be accessed by clicking on the corresponding description card:

- **Configure**: this is the first tool to be used after installation. It configures the **base datasets** and prepares them for publication in the catalogue
- **Upload**: allows users to upload new Soil Data and Metadata from XLSx files including point soil data, project metadata, profile photo metadata, and Laboratory Extra Measure data
- **Point Soil Data**: allows users to view and edit data relating to soil profiles and soil samples
- **Taxonomies**: provides information on all the taxonomies used for point-based soil data
- **Publishing**: enables the preparation, processing, and validation of datasets for end users
- **Hydro PTF2MED**: calculates a basin-scale pseudo-continuous pedotransfer function specifically developed for the Mediterranean region.

Configure

Following installation, MED SIS requires the configuration and publication of the "**base datasets**", which enable the automatic updating of soil data whenever new data is uploaded.

*The "**base datasets**" should be accessible only to data managers and Med-SIS Staff*

The list contains all available **base datasets**, including soil indicators and point-based soil data. These datasets can be configured individually using the "Configure" button or collectively using "Configure all datasets" button.

The configuration is performed on the server and may take several minutes.

The operation may temporarily affect data access during the update process and should therefore be performed with caution.

Refresh List Configure ALL datasets				
Soil Indicator Base Datasets				
Name ↑↓	Abstract	Category ↑↓	Status ↑↓	Actions
Active carbonate (CaCo3)	Finest and most reactive CaCO3 fractions	Chemical Health	Published	
Antimony (Sb)	Antimony content	Chemical Health	To Configure	Configure

Figure 20, the Configure tool

Upload

Data managers use the MED-SIS Back office to store soil data in the database. The primary soil data uploaded to MED-SIS are point-based soil data, which include information on soil profiles and samples. Other types of information required by MED-SIS include: project metadata, photo metadata and laboratory extra measures, i.e. measurements not included in the default data template.

The data must be uploaded in **XLSx format**. A template is available for each type of information in the doc folder of the MED-SIS Github repository (https://github.com/crs4/med_sis/tree/main/src/s4m_catalogue/static/doc).

Specifically, point-based soil data are managed as catalogue resources, form part of the "**base datasets**" and provide the input for the automatic generation of the soil indicators, which constitute another component of the MED-SIS data infrastructure.

For these activities, the process typically starts with an Excel file complaint with one of the templates available for download from the back office.

The *XLS Uploads* button on the back office home page opens the *List* of files that have already been uploaded to the SIS. The list displays the relevant details, including file name, the user who performed the upload, the upload date, the file type, the action performed, and the outcome (Status).

The following actions are available:

- By clicking the  delete icon, the user can remove the upload entry from the list. This does not delete the data imported into the database.
- By clicking the  information icon, the user can open the upload metadata panel
- By clicking "**New XLSx upload**", the user can open the "*New XLSx upload*" panel

The screenshot shows a web interface titled "List of uploads". At the top right is a button "New XLSx Upload". Below it is a search bar and a "Clear filter" button. The main part of the interface is a table with the following columns: Identifier, Name, Editor, Date, Type, Operation, Status, and Actions. The table contains several rows of upload records, including "points_sardegna_monitoring.xlsx" and "points_sicilia_legacy.xlsx". Each row has a status indicator (e.g., "Success") and action buttons (e.g., "X", "B").

Identifier	Name	Editor	Date	Type	Operation	Status	Actions
4	points_sardegna_monitoring.xlsx		08/11/2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Success	X B
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST				Success	X B
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST				Success	X B
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST				Error	X B
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST				Error	X B
5	points_sicilia_legacy.xlsx		08/11/2026				
1	points_andalucia_legacy.xlsx		08/11/2026				
3	points_only_test_jordan_legacy2.xlsx		08/11/2026				
2	points_only_test_jordan_legacy_1.xlsx		08/11/2026				

Figure 21 List of uploaded dataset

An Upload can have one of the following four states:

- **ELABORATING:** the upload is being processed
- **SUCCESS:** the dataset has been successfully uploaded to the server
- **ERRORS:** the dataset has been uploaded to the server, but errors were detected during process,
- **CRITICAL ERROR:** the upload has generated system errors

The screenshot shows a web interface titled "New XLSx upload panel". At the top right is a button "List of uploads". Below it is a "Help" section with a minus icon. The main part of the interface contains instructions for uploading soil data using an XLSx Spreadsheet. The instructions are numbered 1 to 3: 1. First choose the type of data, 2. Second choose the action to perform in the database, 3. Third select the local file to upload. Below the instructions are two dropdown menus: "Choose the type" and "Choose the action", each with a question mark icon. To the right of the dropdowns are two status indicators: a green checkmark indicating "If the pre-validation is successful, you will be able to upload the soil data to the server by clicking on Save button." and a red X indicating "If the data is not valid you can see a report that show errors."

Figure 22 New XLSx upload panel

The *New XLSx upload panel* display several help sections, either directly in the panel by clicking the help icon

The data manager first selects the data type and the action to be performed. The available data type are: Point Soil Data, Laboratory Extra Measure, Project Metadata, and Photo Metadata.

The user then selects the required database action:

- **CREATE IF NOT EXIST** for each item in the uploaded file, MED-SIS attempts to insert it into the database. If an item with the same id already exists, an error is generated.
- **REPLACE IF EXIST**: for each uploaded item, MED-SIS attempts to replace the corresponding item in the database. If the item does not already exist, an error is generated.
- **UPDATE IF EXIST**: for each uploaded item, MED-SIS applies partial changes to the corresponding fields in the database. If an item with the same id does not exist, an error is generated.

For each data type, users can click the question mark icon next to the options list to download the official .XLS template and the instructions for filling it in. Once the options have been selected, the "Choose file" button becomes available.

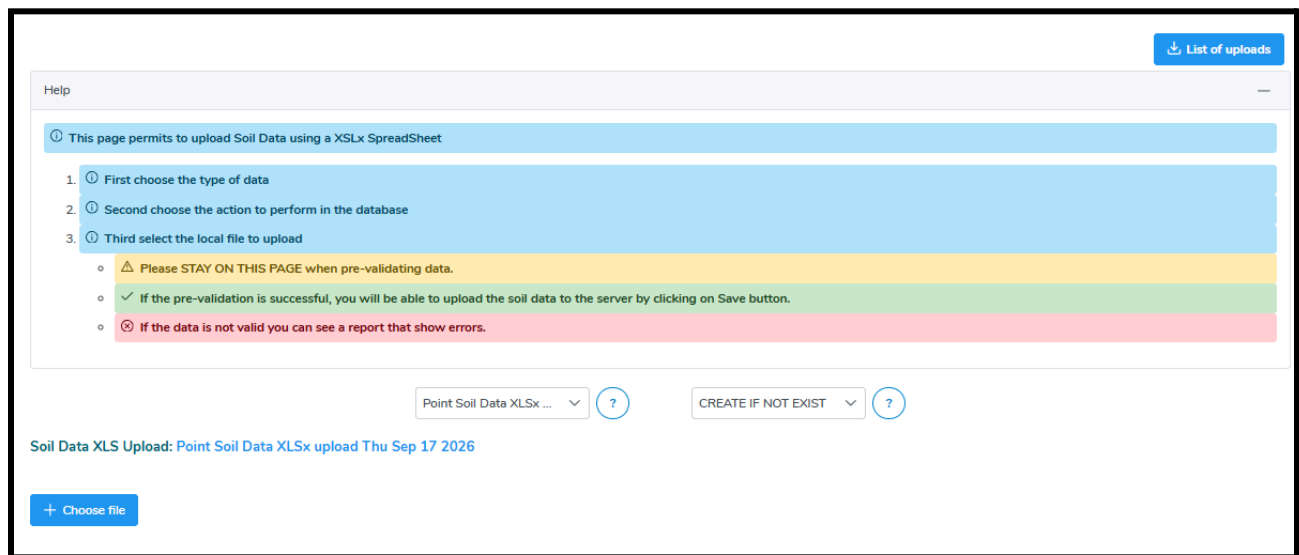
The screenshot shows a web interface for uploading soil data. At the top right is a blue button labeled "List of uploads". Below it is a "Help" section with a list of instructions: 1. First choose the type of data, 2. Second choose the action to perform in the database, and 3. Third select the local file to upload. Below the instructions are three colored boxes: a yellow one with a warning icon saying "Please STAY ON THIS PAGE when pre-validating data.", a green one with a checkmark saying "If the pre-validation is successful, you will be able to upload the soil data to the server by clicking on Save button.", and a red one with an error icon saying "If the data is not valid you can see a report that show errors." Below these are two dropdown menus: "Point Soil Data XLSx ..." and "CREATE IF NOT EXIST", each with a question mark icon. Below the dropdowns is a status message: "Soil Data XLS Upload: Point Soil Data XLSx upload Thu Sep 17 2026". At the bottom left is a blue button labeled "+ Choose file".

Figure 23, Upload tool page

After the XLS file has been successfully uploaded, a confirmation message indicates that the file has been successfully converted to a GeoJSON format. Users can scroll down to view the uploaded points on a map. If no errors occur during the upload, a No Errors message is displayed below the map, as shown in Figure 8.

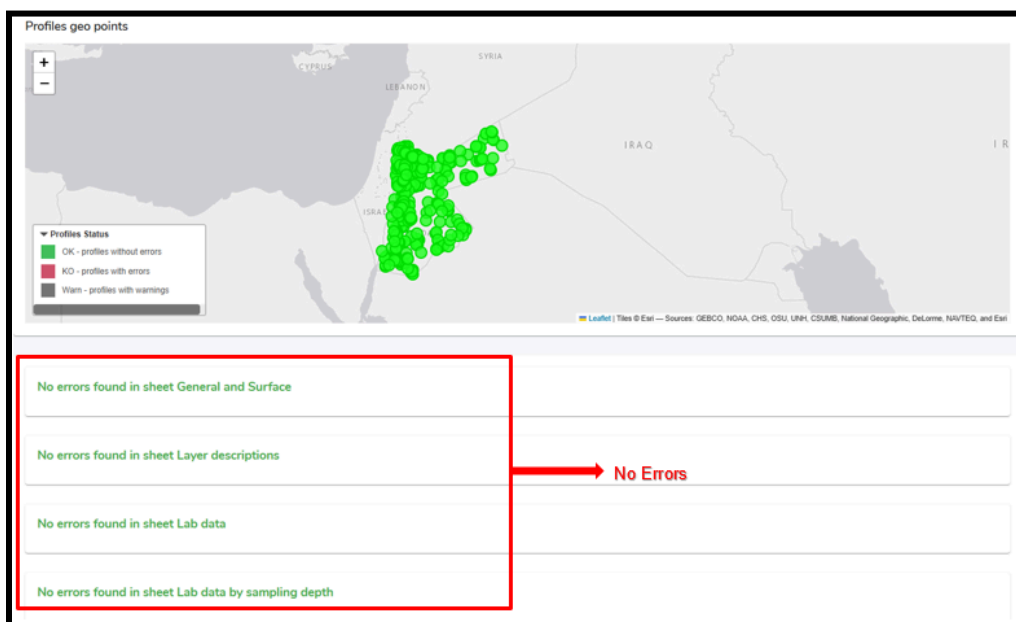


Figure 24 Dataset chosen and errors preview

Users can then: click **Reset** button to restart the upload process, click **Save Data** to upload the data to the database. The Save Data operation may take several minutes to complete.



Figure 25 Dataset chosen and "Save Data" button

If errors occur during the upload, users can view a log reporting the type of error and the corresponding cell (row and column) in the XLSx file.



Figure 25 Dataset chosen with errors

Sheet "Lab data": 9Errors!

▼ Collapse

Element ↑↓	Row ↑↓	Column ↑↓	Error ↑↓
?	111	1 or A	Error: duplicate key found
?	114	1 or A	Error: duplicate key found
?	115	1 or A	Error: duplicate key found
?	116	1 or A	Error: duplicate key found
MED3100@1	463	58 or B5	Wrong Percentage value field
TPER001@2	482	13 or M	Wrong Percentage value field
TPER001@2	482	17 or Q	Wrong Percentage value field
?	483	1 or A	Error: duplicate key found
CARBSKB004@1	537	58 or BF	Wrong Percentage value field

<< < 1 > >> 10 ▼

Figure 26 Errors log

The Point soil data Excel file

According to the *MED-SIS-TEMPLATE_POINT_SOIL_DATA.xlsx* template, the spreadsheet must contain four sheets:

General and Surface

Layer description

Lab data

Soil Classification

The data in General and Surface and Layer description must start from rows 10 and 6, respectively. The data in Lab data and Soil Classification must start from row 3.

Before uploading the file, make sure that all data in the Excel file are stored as 'values' and that no fields contain formulas or text formatting rules. Otherwise, MED-SIS may return errors:

Error Message table

Error msg	Description
Wrong Classification code field	The classification code entered in the cell is not recognized by the SIS.
Wrong Number field	The value entered in the cell is not a number <i>Check whether the cell contains a formula instead of a value</i>
Wrong Boolean field	The value entered in the cell is not a Boolean value. <i>Allowed values: T for True, F for False'</i>
Wrong Date field	The value entered in the cell is not a valid date. The allowed format is DD/MM/YYYY <i>Check whether the cell contains a formula instead of a value</i>
Wrong Percentage value field	The value entered in the cell is not a valid percentage. <i>Check whether the cell contains a formula instead of a value</i>
Wrong Positive value field	The value entered in the cell must be a positive number or 0. <i>Check whether the cell contains a formula instead of a value</i>
Wrong Latitude value field	The latitude must be expressed in decimal degrees. <i>Check whether the cell contains a formula instead of a value</i>
Wrong Longitude value field	The longitude must be expressed in decimal degrees. <i>Check whether the cell contains a formula instead of a value</i>
Error found Rich Text	<i>The cell contains text formatting rules. Please use plain text instead</i>
Error duplicate key found	<i>The element identifier is the same as the identifier of another element in a previous row</i>

Point Soil Data

The Point Soil Data tool allows users to perform CRUD operations (Create, Read, Update, and Delete) on individual soil profiles or samples.

As shown in Figure xxx the first page of the tool displays a list of soil profiles and samples, together with a set of options that allow users to search for and identify specific elements in the list.

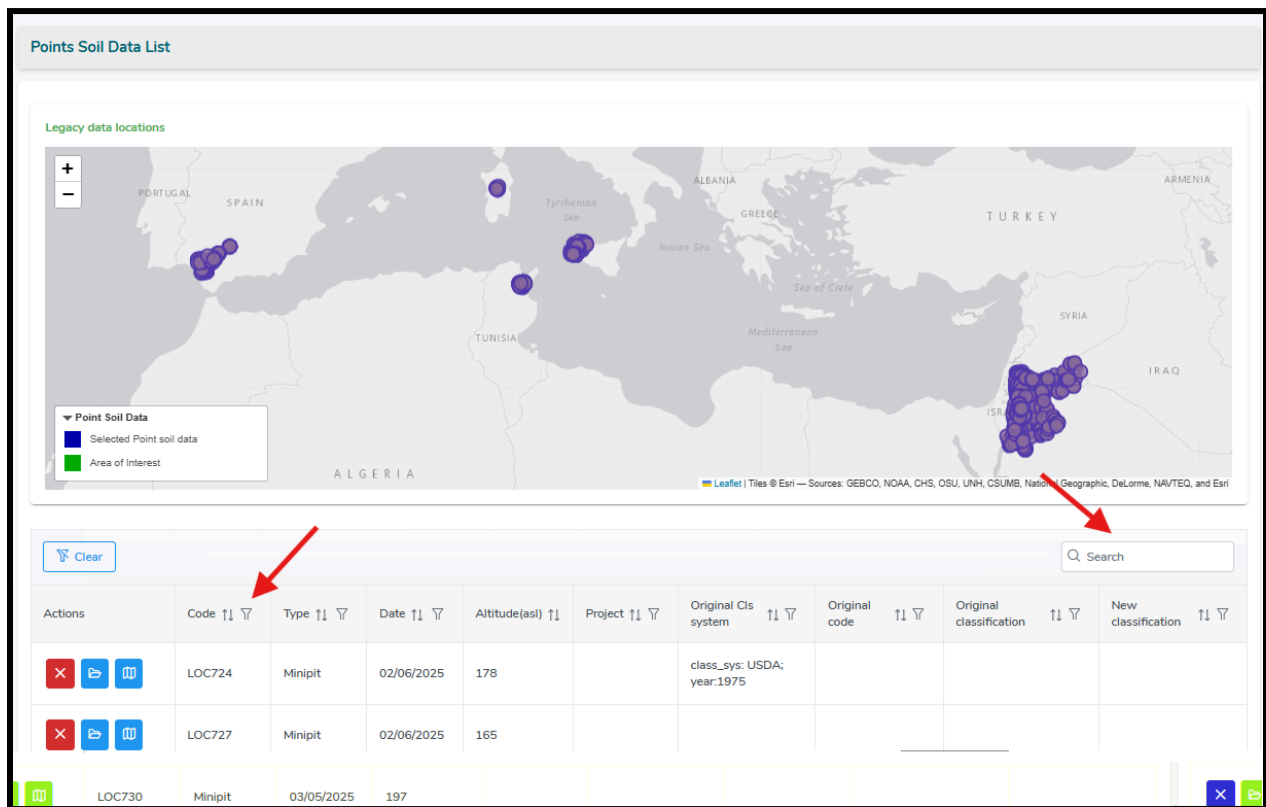


Figure 27 Point Soil Data list

The following actions are available:

By clicking the  delete icon, the user can delete the selected record from the list. This operation also deletes the corresponding imported data.

By clicking the  details icon, the user can open the **Point Soil Data panel** for the selected element.

By clicking the  map icon, the user can display the selected element in the map panel.

The Point Soil Data panel (Figure 28) displays all the information associated with a soil profile or sample.

The red box indicates the summary card describing the profile / sample. The central panel displays the depth and Munsell colour of the profile layers, while the panel on the left shows the profile location.

The red arrow points to the sections containing the profile data. A Profile can potentially contain around one thousand fields.

The fields are organized into sections. Each field has a name, a data type and an editable value.

By clicking the  edit icon, the user can edit the value of the field

By clicking the  information icon, the user can open the information panel associated with the field

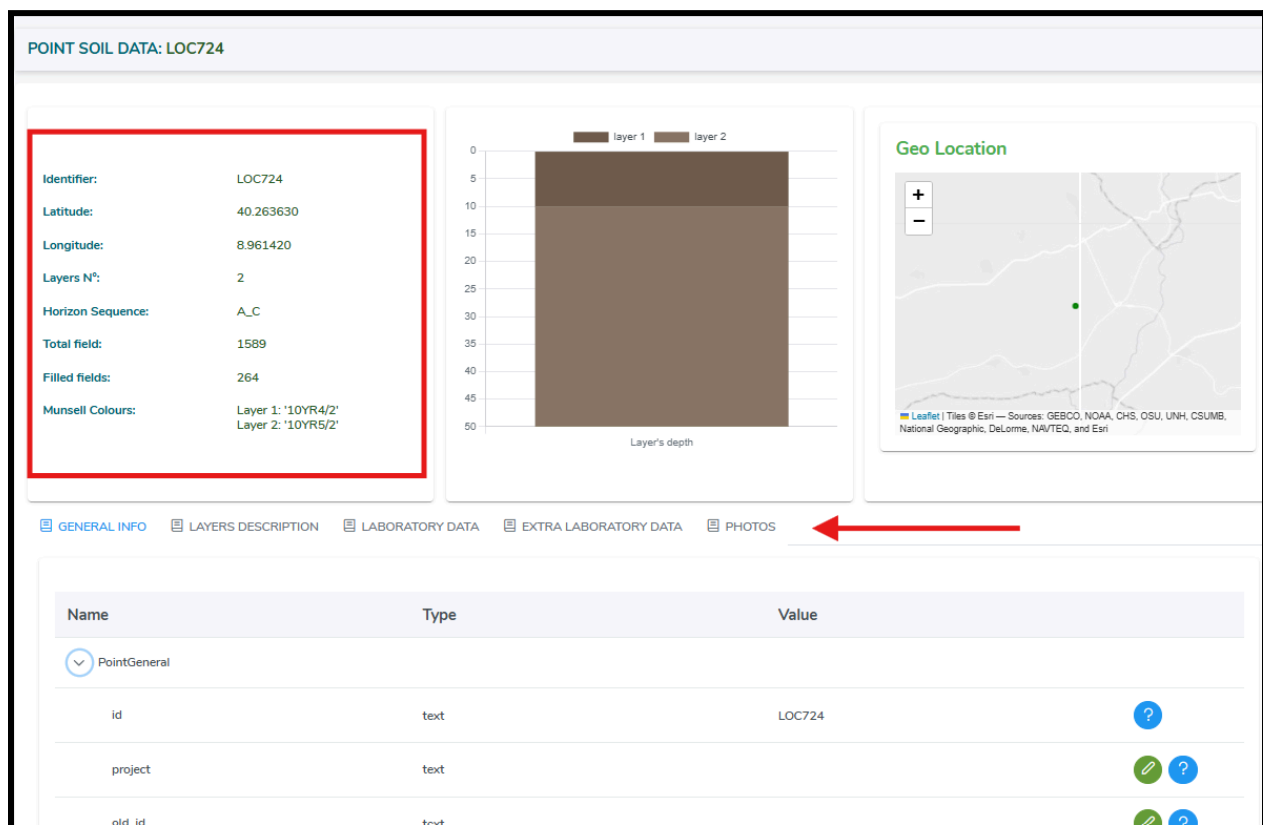
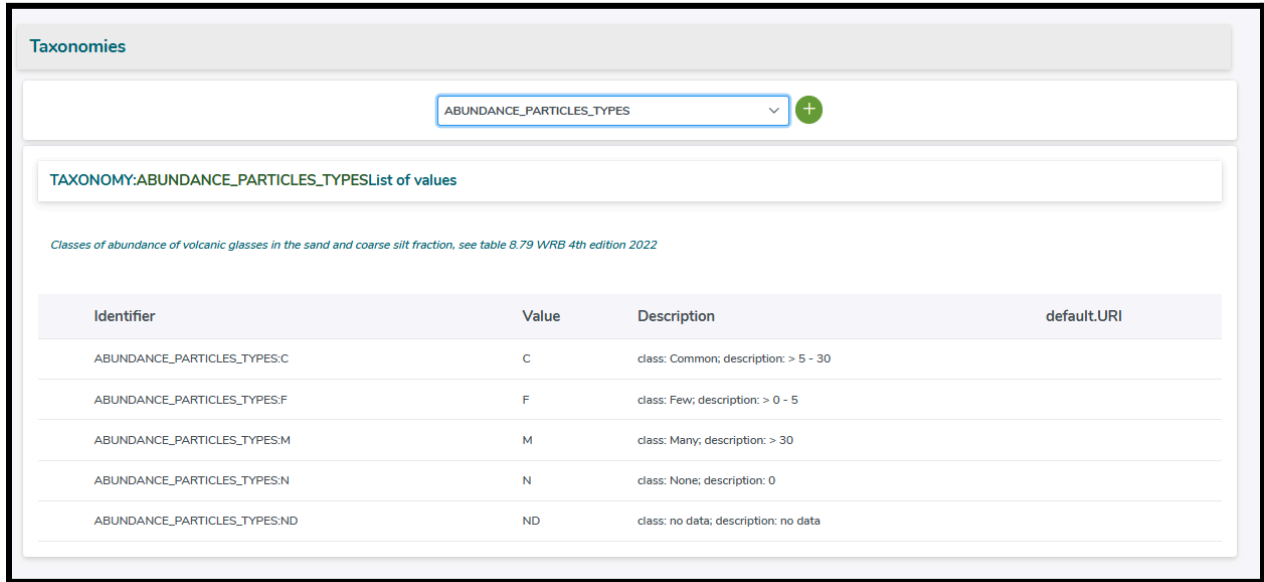


Figure 28 Point Soil Data view

Taxonomies

The Taxonomies tool allows users to explore all the taxonomies used in the creation and management of soil data.



The screenshot shows the 'Taxonomies' tool interface. At the top, there is a header 'Taxonomies'. Below it, a search bar contains the text 'ABUNDANCE_PARTICLES_TYPES' and a green plus icon. A dropdown menu is open, showing a list of taxonomy entries. The first entry is 'TAXONOMY:ABUNDANCE_PARTICLES_TYPESList of values'. Below this, a subtitle reads 'Classes of abundance of volcanic glasses in the sand and coarse silt fraction, see table 8.79 WRB 4th edition 2022'. The main content is a table with four columns: Identifier, Value, Description, and default.URI. The table lists five entries: C, F, M, N, and ND, each with a corresponding description.

Identifier	Value	Description	default.URI
ABUNDANCE_PARTICLES_TYPES:C	C	class: Common; description: > 5 - 30	
ABUNDANCE_PARTICLES_TYPES:F	F	class: Few; description: > 0 - 5	
ABUNDANCE_PARTICLES_TYPES:M	M	class: Many; description: > 30	
ABUNDANCE_PARTICLES_TYPES:N	N	class: None; description: 0	
ABUNDANCE_PARTICLES_TYPES:ND	ND	class: no data; description: no data	

Figure 29 Taxonomy list

The drop-down list allows users to select a taxonomy, which is then displayed as a list. Each item in the list describes a taxonomy entry using: the following fields:

- **identifier:** the identifier used in the database
- **Value:** the name or value of the entry
- **Description:** a description of the entry
- **URI:** an external URI referring to an international standard vocabulary, such as <https://qudt.org/vocab/unit/MilliGM-PER-KiloGM> .

By clicking the  icon, users can open the panel for creating a new taxonomy. This may be necessary when additional laboratory measurements need to be entered.

These operations must be coordinated with the other information systems (SISs) in the network to ensure **interoperability** and data sharing.

Figure 30 Taxonomy create

Once a taxonomy has been created, new entries can be added to it.

Figure 31 Taxonomy - add classification

Taxonomies can also be deleted by clicking the the  **delete** icon.

To ensure interoperability, **default taxonomies cannot be removed or modified.**

Publishing

The *Publishing* tool, accessible from the back office home page, is the MED-SIS **main tool** for create **customized datasets for end user**

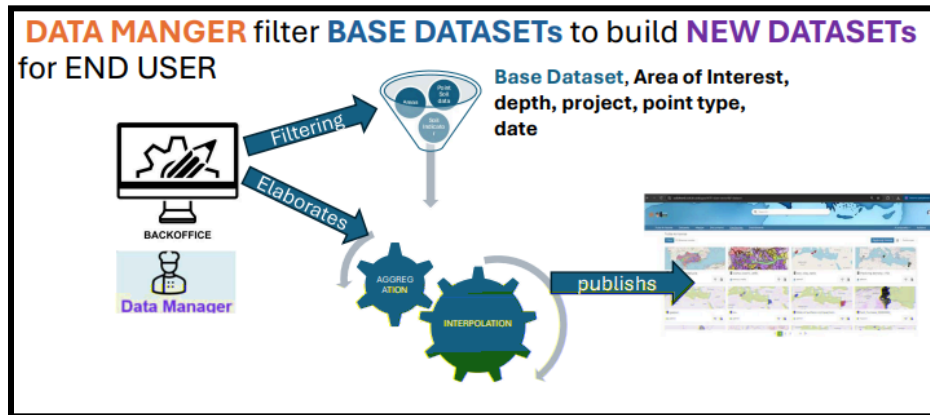


Figure 32 Publishing data flow

The first page displays the *list* of datasets that have already been created, together with relevant details and icons for the available actions.

Dataset List

+ New Dataset

Clear Search

Actions	Code ↑↓	Name ↑↓	Date ↑↓	Context ↑↓	Source ↑↓	User ↑↓	Email ↑↓	Status ↑↓
  	4	admin-dataset-2026-09-10	2026-09-10		Cation Exchange Capacity (CEC) / Clay ratio	admin	demontis@crs4.it	CREATED
  	3	admin-dataset-2026-08-31	2026-08-31		Active carbonate (CaCo3)	admin	demontis@crs4.it	PUBLISHED

<< < 1 > >>

Figure 33 Publishing - customized dataset list

By clicking the  delete icon, the user can remove the dataset entry from the list. **This does not delete the corresponding published datasets from the catalogue.**

By clicking the  details icon, the user can open the dataset panel, which displays the dataset details and current status.

By clicking the “**New Dataset**”, the user can open the “*New Dataset*” panel (Figure 32)

There are three types of customized dataset, depending on the type of source base dataset:

- Point Soil Data Sections: based on a source Base Dataset of type Point Soil Data
- Soil Indicators: based on a source Base Dataset of type Soil Indicator
- Soil indicators evaluated in an area: based on a source Base Dataset source of type Soil Indicator

When a dataset is created, it appears in the dataset list with the status “CREATED”

Choose the context of the new Dataset

You must choose the context of the new dataset

☐ Point Soil Data Sections
☐ Soil Indicators
☐ Soil indicators evaluated in an area

Figure 34 Publishing - customized dataset types

Dataset List

+ New Dataset

Clear

Search

Actions	Code ↑↓	Name ↑↓	Date ↑↓	Context ↑↓	Source ↑↓	User ↑↓	Email ↑↓	Status ↑↓
	4	admin-dataset-2026-09-10	2026-09-10		Cation Exchange Capacity (CEC) / Clay ratio	admin	demontis@crs4.it	CREATED
	3	admin-dataset-2026-08-31	2026-08-31		Active carbonate (CaCo3)	admin	demontis@crs4.it	PUBLISHED
	6	admin-dataset-2026-09-17	2026-09-17			admin	demontis@crs4.it	CREATED
	7	admin-dataset-2026-09-17	2026-09-17			admin	demontis@crs4.it	CREATED

<< < 1 > >>

Figure 35 Publishing - customized dataset types

A dataset can have four possible states:

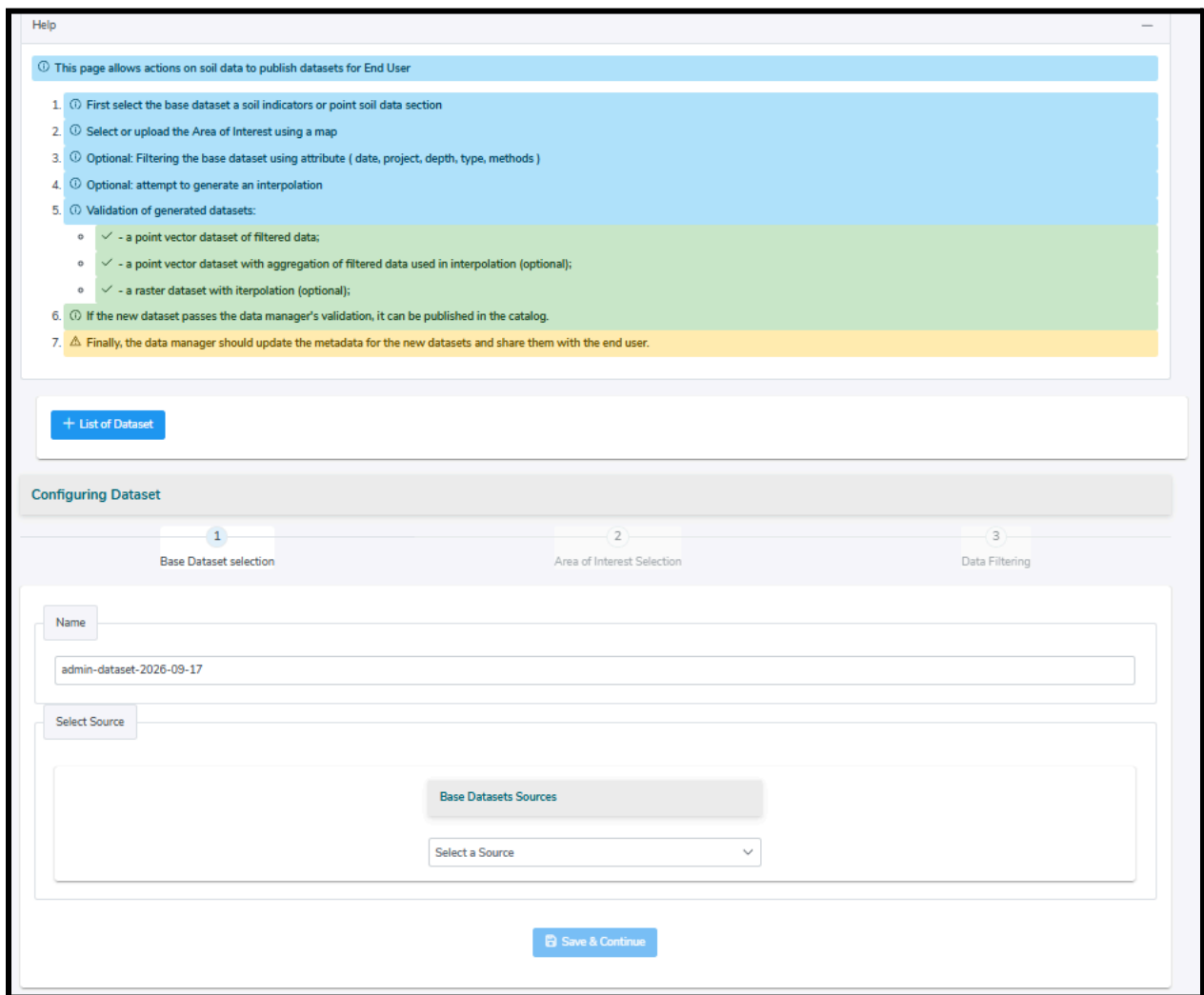
- **CREATED** : the dataset has been created but not configured

- **CONFIGURED** : the dataset has been configured but not published in the catalogue
- **IN_PROCESS** : the dataset is being processed
- **VALIDATED** : the dataset has been validated by the data manger,
- **PUBLISHED** : the dataset has been published in the catalogue,
- **ERRORS** : errors occurred during dataset processing.

The data manager uses the tool to progress the dataset the following workflow :

CREATED => CONFIGURED => IN_PROCESS => VALIDATED => PUBLISHED

By clicking the  details icon the user opens the dataset panel, which displays the current status of the dataset.



Help

① This page allows actions on soil data to publish datasets for End User

- ① First select the base dataset a soil indicators or point soil data section
- ① Select or upload the Area of Interest using a map
- ① Optional: Filtering the base dataset using attribute (date, project, depth, type, methods)
- ① Optional: attempt to generate an interpolation
- ① Validation of generated datasets:
 - ✓ - a point vector dataset of filtered data;
 - ✓ - a point vector dataset with aggregation of filtered data used in interpolation (optional);
 - ✓ - a raster dataset with interpolation (optional);
- ① If the new dataset passes the data manager's validation, it can be published in the catalog.
- ⚠ Finally, the data manager should update the metadata for the new datasets and share them with the end user.

+ List of Dataset

Configuring Dataset

1 Base Dataset selection 2 Area of Interest Selection 3 Data Filtering

Name

admin-dataset-2026-09-17

Select Source

Base Datasets Sources

Select a Source

Save & Continue

Figure 36 Publishing - dataset configuration

Figure 37 Publishing - Configuration Step 1. Base Dataset Selection

After selecting the source **Base Dataset**, the system loads **all the points** contained in the selected Base Dataset. The user can then click **Save & Continue**.

Figure 38 Publishing - Configuration Step 1. Base Dataset Selected report

The next step is to select the Area of Interest (AOI). The AOI can be defined either by drawing a custom polygon (Figure 35) or by selecting polygons from a Boundaries Catalogue dataset (Figure 36)

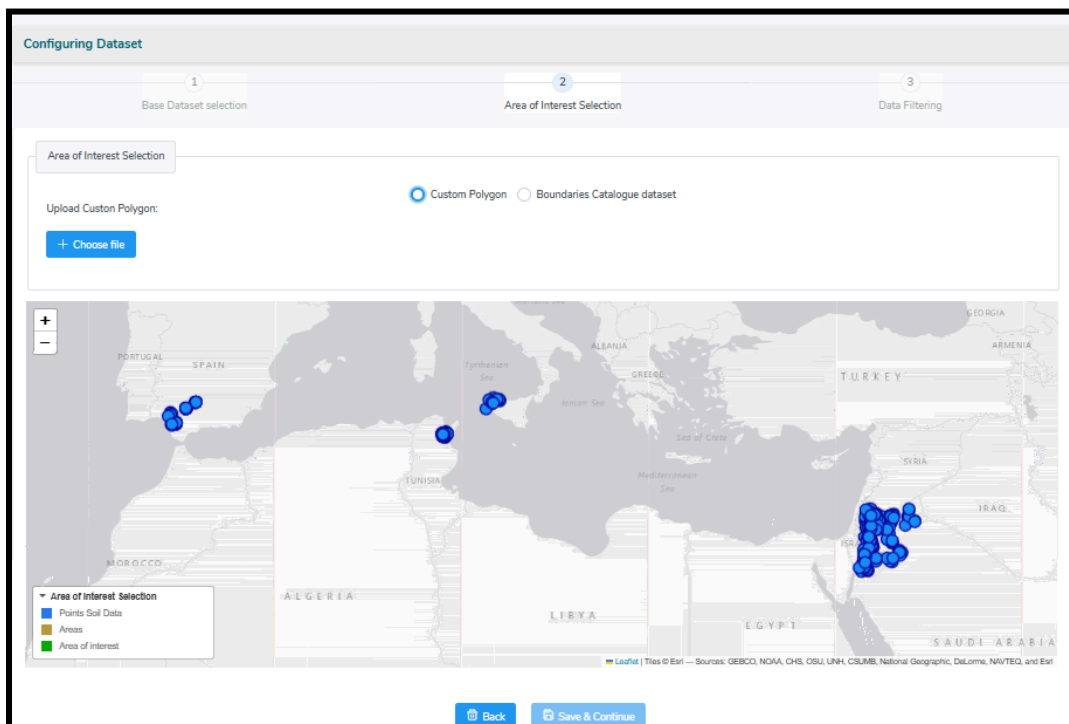


Figure 39 Publishing - Configuration Step 2. Custom polygon as AOI

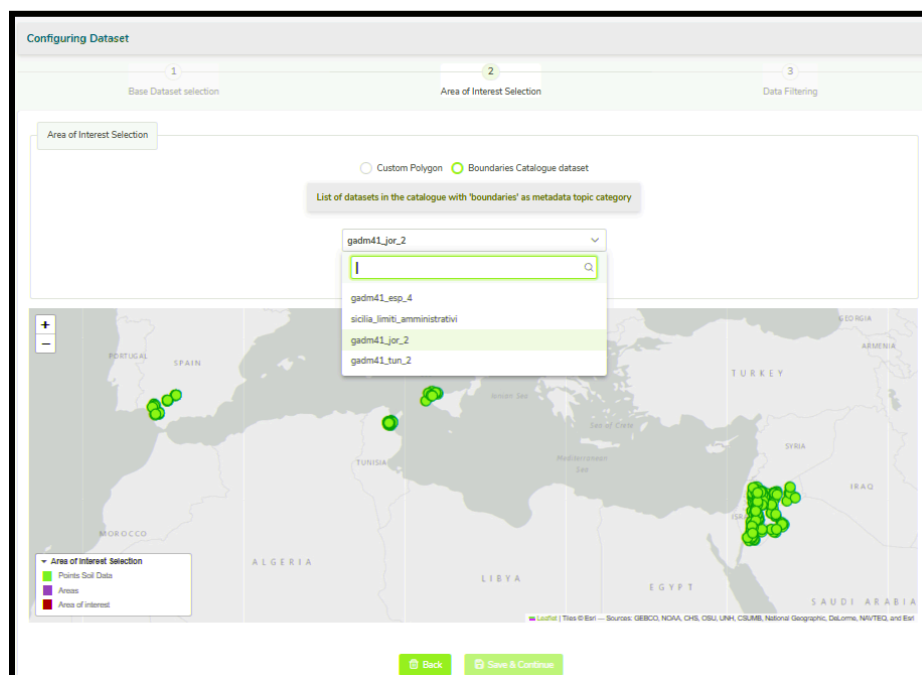


Figure 40 Publishing - Configuration Step 2. Catalogue resources as source for AOI

To be used as "Boundaries" dataset, a catalogue dataset must have its "Topic category" metadata field set to "Boundaries".

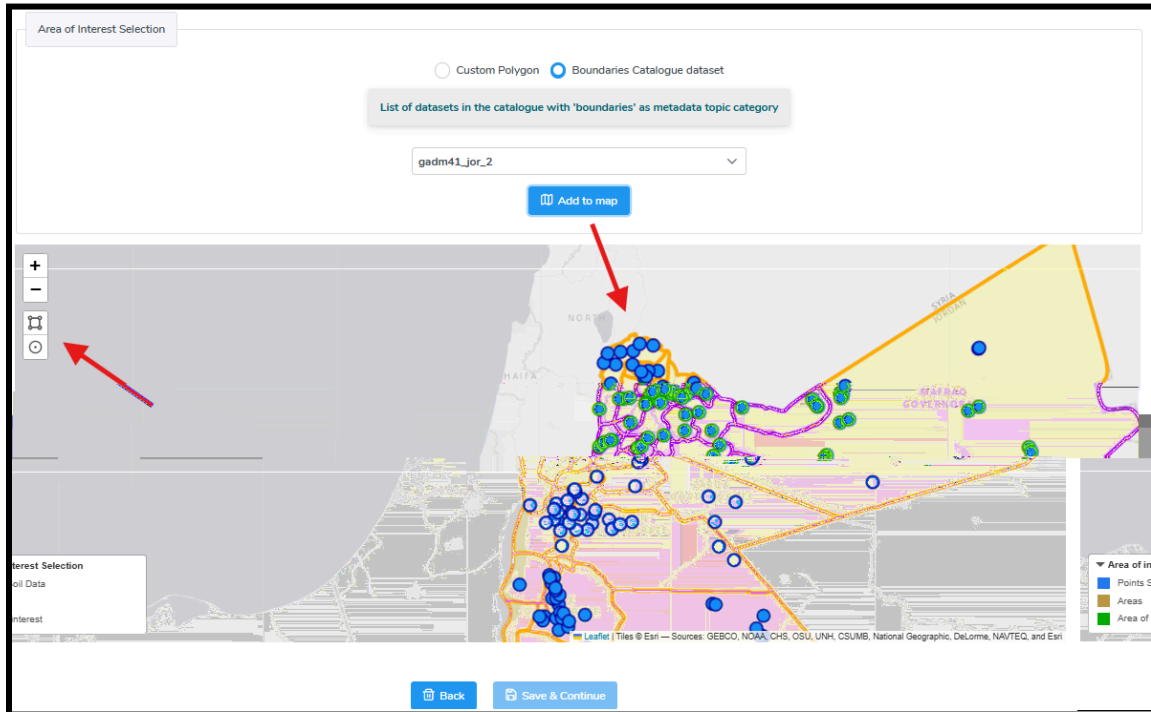


Figure 41 Publishing - Configuration Step 2. Catalogue add source for AOI

After adding the dataset to the map, users can select the required polygons using a point or a selection box, after activating the corresponding tool in the top-left corner of the map.

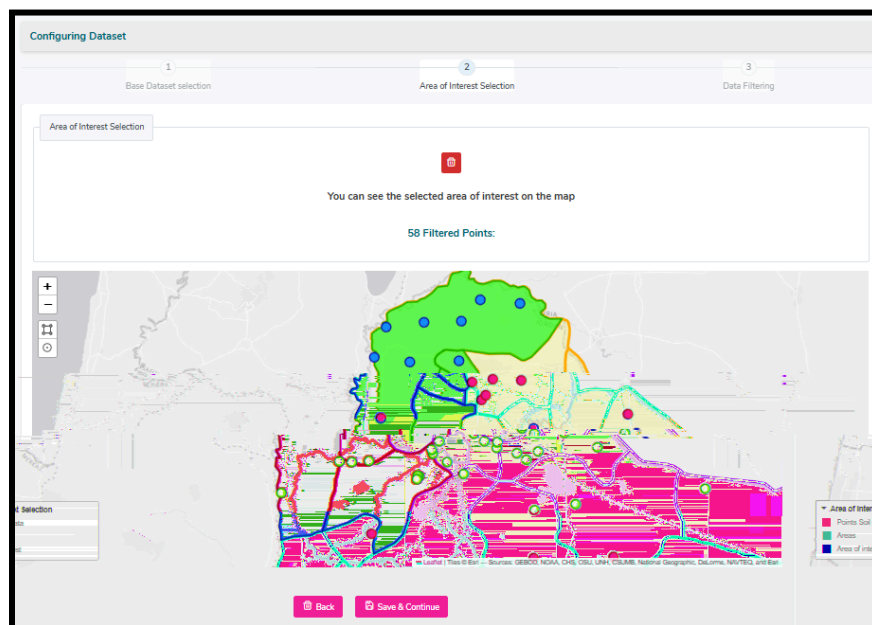


Figure 42 Publishing - Configuration Step 2. AOI selection

After the AOI has been selected, the number of filtered points is displayed on the page. The user can then click “Save & Continue”.

The next configuration step consists of filtering the points according to selected attributes s:

- **observation period** start and end dates, or all dates
- **type** : *Auger hole, Minipit or Profile*, or all types
- **project**: the Identifier or name of the project, survey activity/campaign, within which the point based soil data were collected or all projects
- The **depth**: upper and lower depth, where available. User can select the 0- 20 cm interval , the 20 to 50 cm interval , or all depths
- **laboratory method**: the method used for the measurements, no method or all methods. This filter is available depending on the measurements involved and applies only to soil indicators.
- **the field survey** method: the method used to carry out field survey, or all methods.

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

Figure 43 Survey Methods

The filtering panel (Figure 41) displays a form for applying the filters and a map for visualizing the result.

The information at the top of the page indicates the number of selected points selected and the number of aggregated points required for interpolation. The aggregation is based on the location of the points, using a 10m point buffer and on a weighted average of values based on depth intervals (for soil indicators only). The values available in the drop down lists correspond to the data contained in the selected points.

After applying the filters, the user can click “Save & Continue”.

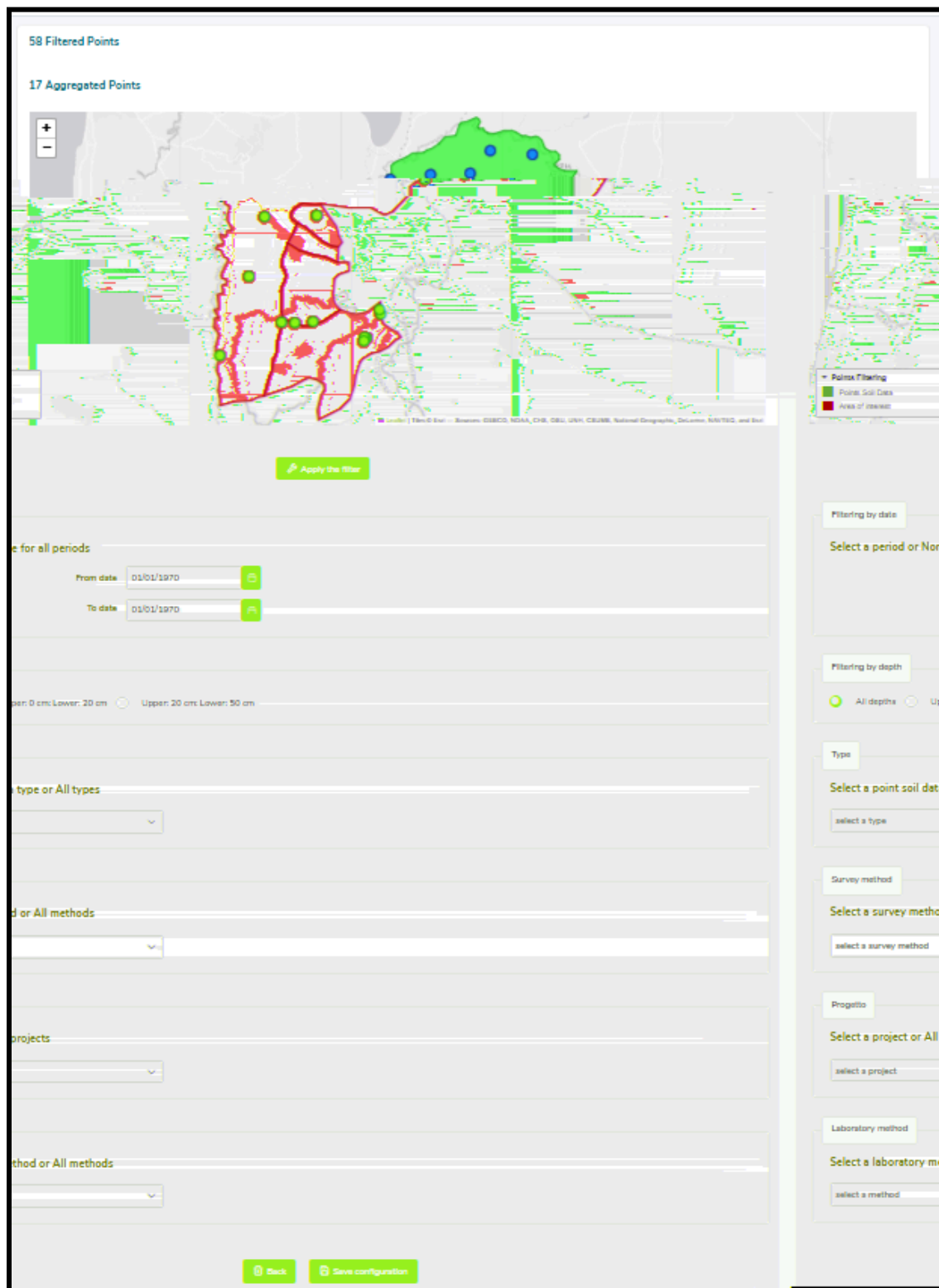


Figure 44 Publishing - Configuration - Filtering panel

The customized dataset then enters the validation phase.

The system displays a report showing the number of filtered and aggregated points and indicating whether the dataset contains **outliers**.

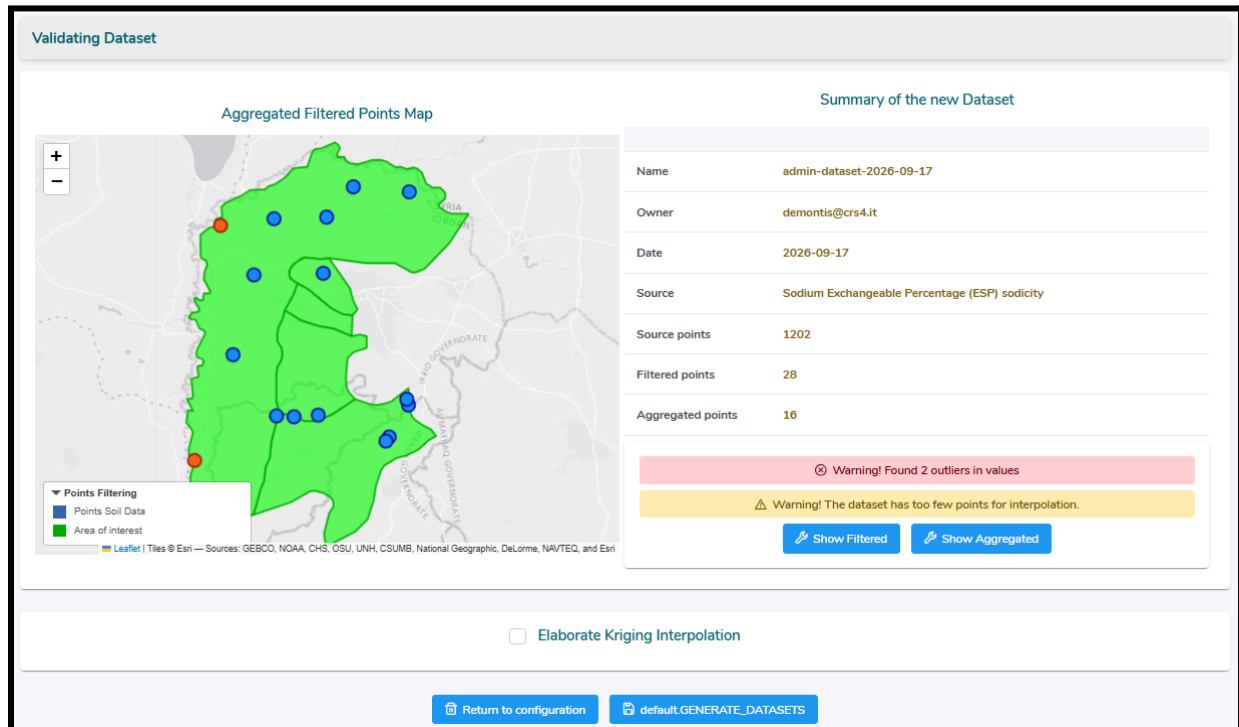


Figure 45 Publishing - Validation - Report panel

The presence of outliers may result in predictions outside the expected range by skewing parameter estimates and **altering the underlying data distribution**.

Outliers can result from data entry errors, but they can also represent genuine data points that WARRANT FURTHER INVESTIGATION.

If the dataset does not require further processing, such as Kriging interpolation, it can be published directly in the catalogue.

Otherwise, selecting the "Elaborate Kriging Interpolation" checkbox opens the Kriging interpolation panel (Figure 43).

In this panel, users can set the interpolation parameters and evaluate the resulting semivariogram (Figure 44)

If the validation results are satisfactory, the data manager can proceed with the generation of the new datasets.

Kriging Interpolation

Select Kriging interpolation parameters

☐ Do not use outliers

The presence of outliers generates out of scale predictions by skewing parameter estimates and altering the underlying data distribution. Outliers can result from data entry errors, but they can also be genuine data points that WARRANT MORE INVESTIGATION.

Model (Default: linear " $a + x * b$ ")

Linear

$a + b * x$

Initial Number of Distance Classes (Minimum: 2, Default: 100)

-

100

+

Maximum Distance Floating point (Default: -1.000000, not used)

-

-1

+

Skip Number (Minimum: 1, Default: 1)

-

1

+

Calculate the variogram

Return to configuration

default.GENERATE_DATASETS

Figure 46 Publishing - Validation - Kriging Interpolation

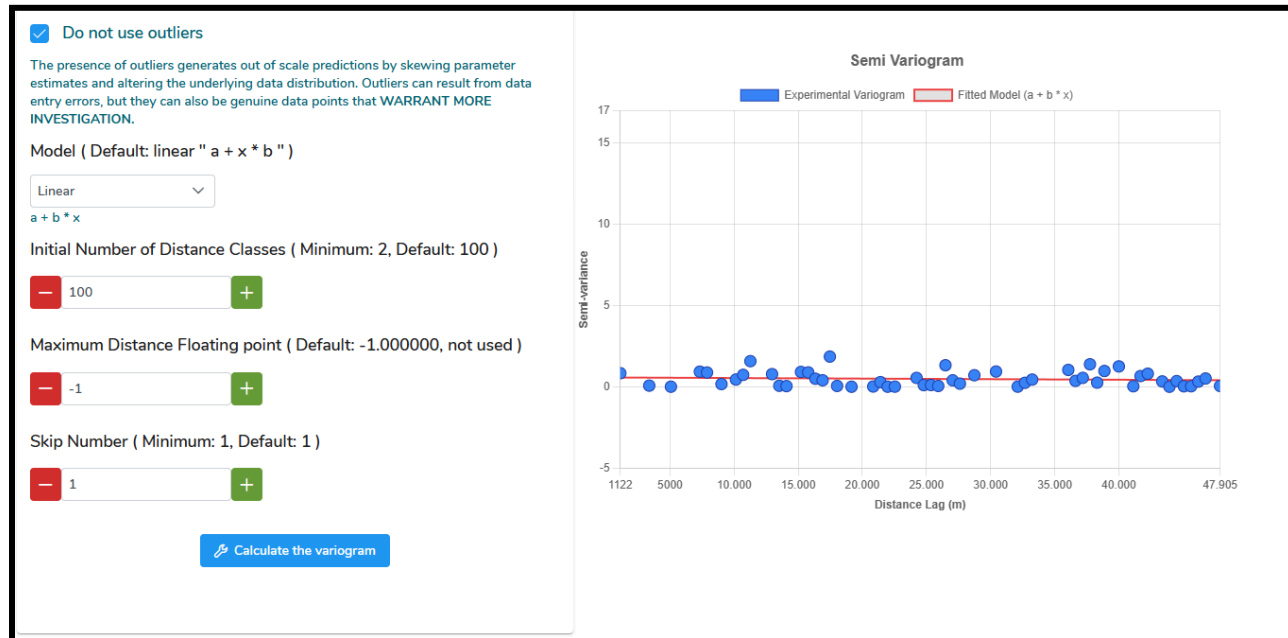


Figure 47 Publishing - Validation - Kriging Interpolation - Semi variogram

Dataset List

+ New Dataset

Clear

Actions	Code	Name	Date	Context	Source	User	Email	Status
 	6	admin-dataset-2026-09-17	2026-09-17	 Sol Indicator	Sodium Exchangeable Percentage (ESP) sodicity	admin	demonits@crs4.it	IN_PROCESS
 	4	admin-dataset-2026-09-10	2026-09-10	 Sol Indicator	Cation Exchange Capacity (CEC) / Clay ratio	admin	demonits@crs4.it	CREATED

Figure 48 Publishing - Datasets in elaboration

Dataset List

+ New Dataset

Clear Search

Actions	Code	Name	Date	Context	Source	User	Email	Status
	6	admin-dataset-2026-09-17	2026-09-17	Soil Indicator	Sodium Exchangeable Percentage (ESP) sodicity	admin	demonitis@crs4.it	PUBLISHED
	4	admin-dataset-2026-09-10	2026-09-10	Soil Indicator	Cation Exchange Capacity (CEC) / Clay ratio	admin	demonitis@crs4.it	CREATED

Figure 49 Publishing - Datasets published

After processing, the customized dataset is published in the catalogue. The publishing report proposes a style (Figure 46) for the raster dataset because, since the generated raster dataset is assigned a default style

Owner	demonitis@crs4.it
Date	2026-09-17
Source	Sodium Exchangeable Percentage (ESP) sodicity
Source points	0
Filtered points	28
Aggregated points	16
No results found	

Proposed Raster Style:

```
<?xml version="1.0" encoding="UTF-8"?>
<StyledLayerDescriptor version="1.0.0"
  xsi:schemaLocation="http://www.opengis.net/sld StyledLayerDescriptor.xsd"
  xmlns="http://www.opengis.net/sld" xmlns:ogc="http://www.opengis.net/ogc"
  xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  <NamedLayer>
    <Name>prediction_raster</Name>
    <UserStyle>
      <Title>Raster of prediction</Title>
      <Abstract>Raster of prediction</Abstract>
      <FeatureTypeStyle>
        <Rule>
          <Name>Prediction_raster</Name>
          <Title>Prediction raster</Title>
          <RasterSymbolizer>
            <ChannelSelector>
              <GrayChannel>
                <SourceChannelName>1</SourceChannelName>
                <ColorMapEntry>
                  <ColorMapEntry label="1.7533318996429443" quantity="1.7533318996429443"
                    color="#f4a661"/>
                  <ColorMapEntry label="1.8322239518165588" quantity="1.8322239518165588"
                    color="#fdae61"/>
                  <ColorMapEntry label="1.9111160039901733" quantity="1.9111160039901733"
                    color="#fdae61"/>
                  <ColorMapEntry label="2.0560301081861823" quantity="2.0560301081861823"
                    color="#fdae61"/>
                  <ColorMapEntry label="2.2665091463314623" quantity="2.2665091463314623"
                    color="#fdae61"/>
                  <ColorMapEntry label="2.4999999999999998" quantity="2.4999999999999998"
                    color="#fdae61"/>
                </ColorMapEntry>
              </GrayChannel>
            </ChannelSelector>
          </RasterSymbolizer>
        </Rule>
      </FeatureTypeStyle>
    </UserStyle>
  </NamedLayer>
</StyledLayerDescriptor>
```

Figure 50 Publishing - Report

To change the default raster style, open the dataset in the catalogue and select “Edit” menu then “Edit style”

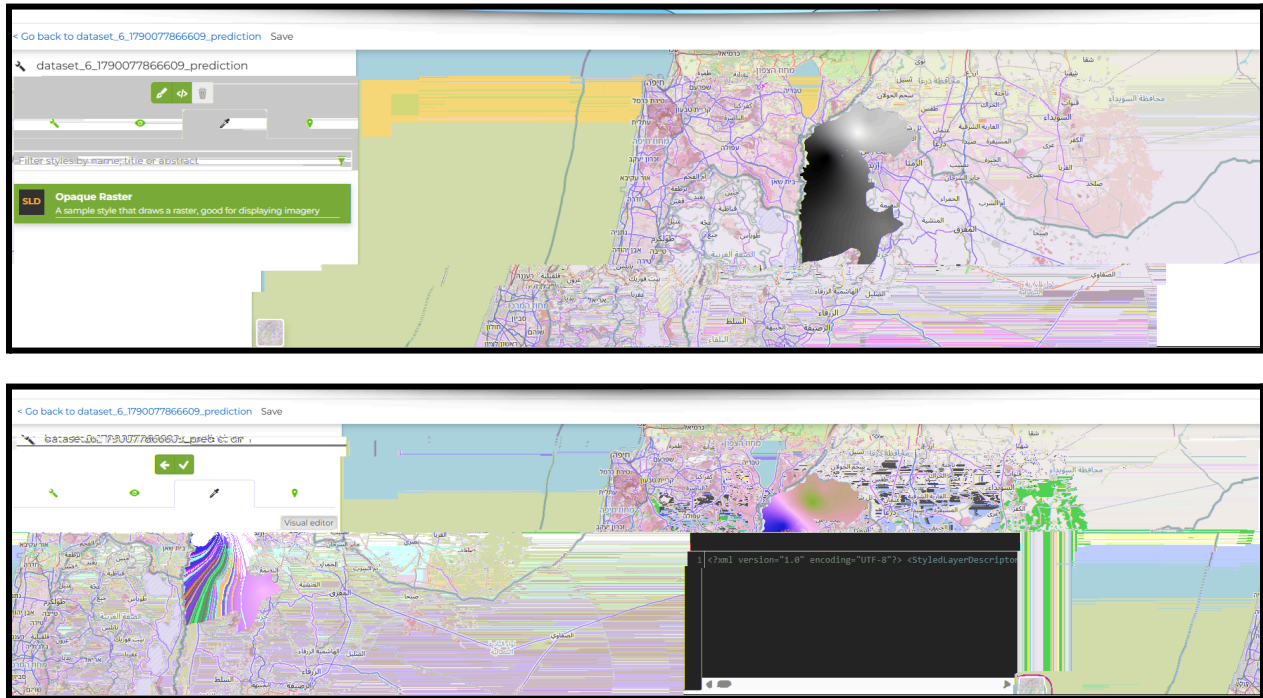


Figure 51 Publishing - changing raster style

Finally, data managers **must edit the metadata and change the name** of the generated datasets in the catalogue. They must then assign the appropriate permissions to the End User allowing them to view and download the datasets.

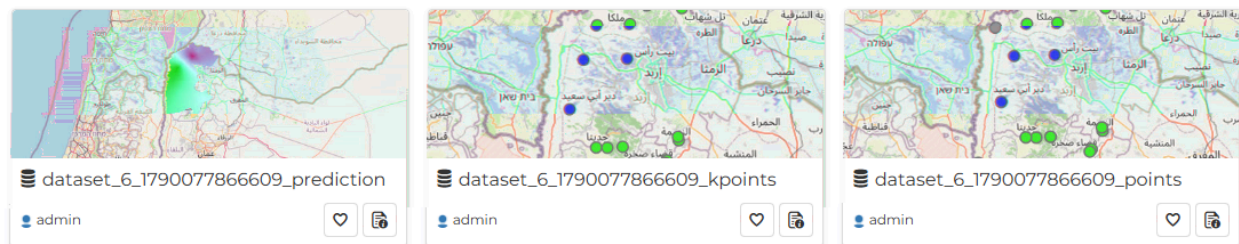


Figure 52 Publishing - generated rasters

The customized dataset generated by the publishing tool are:

- raster dataset containing the interpolation results (optional)
- point vector dataset of aggregated points, whose name ends with '_kpoints'. This dataset can be used to perform further interpolation in desktop GIS software.
- point vector dataset of filtered points, whose name ends with '_points'.

